

Privee: A Wearable for Real-Time Bladder Monitoring System

Ruoyu Zhang
University of California, Davis
Davis, California, USA

Ruijie Fang
University of California, Davis
Davis, California, USA

Chongzhou Fang
University of California, Davis
Davis, California, USA

Houman Homayoun
University of California, Davis
Davis, California, USA

Gozde Goncu-Berk*
University of California, Davis
Davis, California, USA

ABSTRACT

Urinary incontinence (UI) is a prevalent condition affecting millions of individuals worldwide, leading to various physical, social, and psychological challenges that diminish their quality of life. Current management approaches primarily focus on containment rather than proactive monitoring and warning systems. This paper presents the development and evaluation of a novel wearable technology called Privee, designed as an unobtrusive undergarment to monitor bladder fullness in real-time. Privee utilizes e-textile-based bioimpedance spectroscopy technology, which noninvasively assesses bladder fullness by analyzing the electrical properties of body tissues and fluids. The undergarment incorporates eight embroidered electrodes and textile transmission lines seamlessly integrated into the fabric. By continuously monitoring the bioimpedance signals from the bladder, Privee provides real-time information about the bladder's fullness level. This data is processed using a specialized algorithm to estimate the need for urination. The noninvasive nature of Privee eliminates the discomfort and risks associated with invasive monitoring methods, offering a user-friendly and convenient solution for individuals with UI, overactive bladder, or post-operative care needs. This innovative technology has the potential to improve patients' quality of life and optimize healthcare costs associated with UI management.

CCS CONCEPTS

• Applied computing → Consumer health.

KEYWORDS

Internet of Things(IoT), Bio-Impedance Measurement, Wearable Device, Non-Invasive, Bladder Real-Time Monitoring

ACM Reference Format:

Ruoyu Zhang, Ruijie Fang, Chongzhou Fang, Houman Homayoun, and Gozde Goncu-Berk. 2023. Privee: A Wearable for Real-Time Bladder Monitoring System. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*, October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3594739.3610782>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0200-6/23/10.

<https://doi.org/10.1145/3594739.3610782>

1 INTRODUCTION

Urinary incontinence (UI) is defined as involuntary urination or loss of bladder control that leads to social, psychological and hygienic problems that decrease quality of life [2]. According to the statistics of “National Association of Incontinence” over 200 million people suffer from UI worldwide due to different reasons such as menopause, obesity, vaginal delivery, past hysterectomy or Alzheimer's, Dementia, Parkinson's diseases as well as Cerebral Palsy or spinal cord injury that can negatively affect awareness of the bladder control [21]. Incontinence can have major impacts on patients' quality of life and their healthcare costs. Currently, management regimens for UI focus on containing the urine rather than monitoring and warning about the involuntary urination [1]. In this study, we present development and testing of a novel, e-textile based, real-time bladder monitoring wearable technology, Privee, in the form factor of an unobtrusive undergarment worn around the bladder region (Figure 1). Privee is a wearable undergarment that monitors bladder fullness noninvasively without the need for catheterization using bioimpedance spectroscopy technology. It incorporates eight embroidered electrodes and textile transmission lines, which are integrated into the fabric of the undergarment. Bioimpedance spectroscopy technology works by utilizing the electrical properties of body tissues and fluids to assess bladder fullness. By continuously monitoring the bioimpedance signals from the bladder, Privee can provide real-time information about the bladder's fullness level. This data is analyzed and interpreted using a specialized algorithm to estimate the need to urinate. The noninvasive nature of this technology eliminates the discomfort and potential risks associated with invasive bladder monitoring methods. It offers a convenient and user-friendly solution for individuals who require regular monitoring of their bladder function, such as those with urinary incontinence, overactive bladder, or post-operative care needs.



Figure 1: Front and Back View of Privee



Figure 2: Front and Back on Human Body and PCB to E-textile Sensing System Connection

2 BACKGROUND

Scholars have attempted to develop technologies for measuring the urine volume in the bladder using non-invasive modalities, such as ultrasound [5], near-infrared spectroscopy (NIRS) [19] and ultra-wideband (UWB) radar [17]. However, these technologies either rely on relatively big and heavy devices which make it difficult to embed in a wearable form factor for real-time monitoring or are very expensive. Among recent literature on bladder monitoring, using bio-impedance analysis has demonstrated reliable outcomes [16]. Bio-impedance reflects how much a body impedes electric current. Impedance, denoted Z , is an expression of the opposition that an electronic component, circuit, or system offers to alternating and/or direct electric current [23]. Impedance measurement is the simple concept of measuring the electrical current passing through a sample at various frequencies of the excitation voltage. Specifically, bioimpedance spectroscopy is the technique that measures tissue conductivity in response to a small bio-electrical current [4]. While the current is conducted well by water and electrolyte-rich tissues such as the blood and muscle, it is poorly conducted by fat, bone, and air-filled spaces [15]. Bio-impedance is a non-invasive signal that has many clinical and emerging applications in numerous domains including biologic tissue such as monitoring hydration, and body composition [20] lymphedema [24], volume status in dialysis patients [22], cardiac function [3] or lung volume [18]. The bio-electrical current can be applied over a range of frequencies, and impedance to the flow of the current can be quantified and used to determine fluid volumes in the body including the bladder [7]. With the development of real-time monitoring wearable devices [6, 26], A wearable device that could signal bladder volume continuously would be of tremendous medical benefit. However, there exists no practical wearable solution for patients with UI. We introduce development and human subject testing of a soft wearable undergarment with reusable and washable e-textile sensing technology that look and feel like everyday clothing along with a novel low-cost embedded control system and a machine learning algorithm. The proposed solution is expected to improve lives of people with UI and their caregivers with an inexpensive, ubiquitous, non-invasive, non-intrusive everyday e-textile based wearable technology. Beyond the dramatic changes in body awareness and quality of life, the potential for lowering the incidence of urine infection and kidney disease would save substantial healthcare costs.

3 DESIGN STATEMENT

The design process of Privee was guided by a framework of design requirements developed through face to face interviews with 10 UI patients to understand patients' problems associated with UI and to determine needs related to a potential UI wearable technology device [13]. We focused on limiting patients' dependence on absorbent products, avoiding involuntary urination and enhancing psychosocial health through an intimate user experience. Making management of incontinence as invisible as possible, comfortable fit during static and dynamic conditions, ease of donning and doffing and data accuracy were determined as some of the important design parameters for Privee.

Privee is designed to be worn under an underwear so that it makes constant skin contact (Figure 2). The crotch area of the wearable is intentionally left open to avoid donning/doffing during urination so that the precise location of the wearable on the body can be maintained. After testing various belt forms, Privee is designed with legs which limit vertical movement of the wearable on the body. Privee features adjustable extensions from the side seam that wrap around the lower abdomen and attach to a hook/loop fastener at the center back. The extensions provide extra stabilization and constant pressure on the e-textile electrode array ensuring stable electrode/skin contact. It also helps to adjust the fit to the wearers comfort level. Privee houses an e-textile sensing system embroidered on the inner lining which connects to the hardware in a 3D printed case via metal snaps. Two silicone strips are sewn on the waistline and below the e-textile electrode array to minimize movement. Accurate placement of Privee on the body is also crucial for reliable data acquisition. It should be positioned the same way on the body every time it is worn to ensure reliable bioimpedance data therefore a visual landmark is created that allow the wearer to align it on the navel.

3.1 The E-textile Sensing System

Privee's e-textile electrodes and signal transmission lines joining electrodes to the hardware are realized by directly embroidering silver coated, polyamide core conductive thread onto the wearable fabric [14]. We developed the e-textile electrodes using the Tajima



Figure 3: E-textile Sensing System

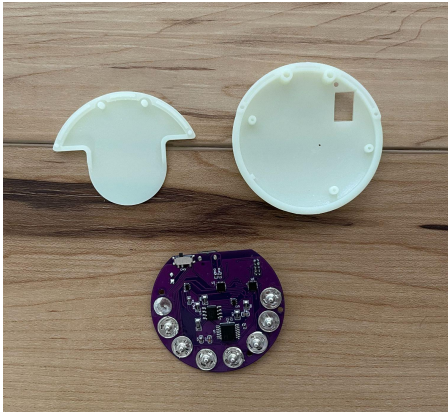


Figure 4: PCB and 3-D Printed Case

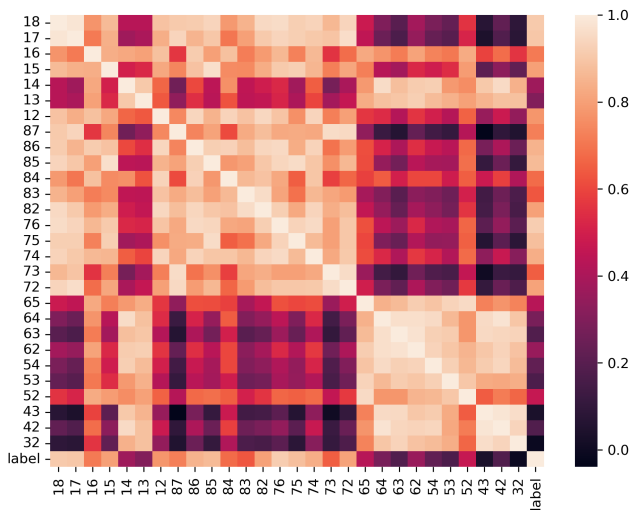


Figure 5: Correlation heat map of testing results

DJ 16 by Pulse software and a Barudan CAD embroidery machine (Figure 3).

We employed an 8-electrode array which together with the e-textile transmission lines are strategically placed to create a sensing network around the abdomen area, allowing for precise and accurate measurements. The embroidered electrodes serve as sensors, detecting and measuring the impedance or resistance to electrical currents that occur as the bladder fills with urine. The textile transmission lines provide the necessary connectivity between the electrodes and the monitoring system, enabling the transmission of data and signals.

3.2 Design of Electronics Hardware

The bio-impedance data is collected and transmitted to computer via a wearable electronics hardware (Figure 4) modified by our previous work[25]. The main Micro-controller Unit (MCU) we use is nRF52832 (Nordic, Inc) which is combined with the Bluetooth Low Energy (BLE) module and the Arm Cortex-M4 CPU together as a wireless multiprotocol System-on-Chips (SoCs). The other main chip adopted is AD5933 (Analog Devices, Inc) high-precision impedance converter system. This chip uses the sine wave signal to detect the impedance at specific frequency. In this case, we use the frequency of 50k based on the literature[12]. Privee employs a 8-electrode array and we measure the impedance within either of the two electrodes in the array. Thus, we use two 8-to-1 Multiplexers (ADG1608 Analog Device, Inc) to choose the either two electrodes and connect them to the AD5933. In the meanwhile, because of the demand for the calibration, we also use two 2-to-1 Multiplexers (TS5A4624 Texas Instruments Inc) to connect a 1k ohms high precision resistor (Panasonic, Inc) into AD5933. As for the power management, TP5400 chip (NanJing Top Power ASIC Corp) fulfills the battery charging, steady voltage providing functions. A customized 50 mm diameter circular PCB board is developed with built in metal snaps for attachment to the e-textile sensing system (Figure 4). In the PCB design, gold immersion is employed to increase the conductivity of the surface between the PCB and the metal snaps. In addition, a case is designed to house the PCB and 3-D printed in Acrylonitrile Butadiene Styrene (ABS) which allows heavy duty use.

4 PILOT USER TESTING

4.1 Design of the Machine Learning Algorithm

Machine learning has seen rapid development and achievement in data analysis in the past decades [8, 10]. We conducted a within-subject machine learning analysis to thoroughly assess the efficacy of Privee. For features, we computed the difference between two cross-sectional impedance values which yielded 378 ($28 \times 27/2$) different features. By calculating these differences, we aimed to mitigate the influence of factors like sweating, ensuring classification solely depends on pre- and post-urination. To facilitate classification, we assigned binary labels (1:pre urination and 0:post urination) to the data. To discern patterns and make predictions, we utilized a support vector machine (SVM) with a “rbf” kernel [9, 11].

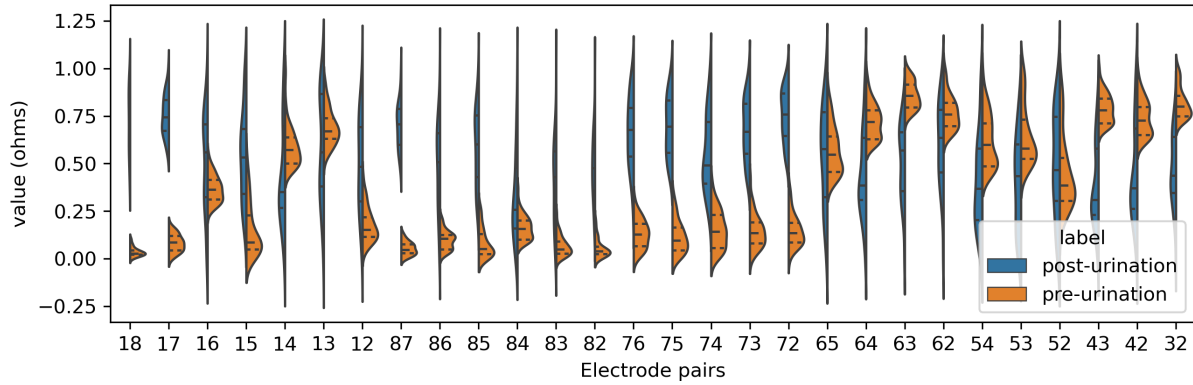


Figure 6: Violin plot of testing results (label 0:post urination, label 1:pre urination)

We conducted a 3-times urination data collection study involving a female participant, focusing on data acquisition before and after urination. The experimental protocol comprised two distinct phases: Phase 1 involved data acquisition before urination, while Phase 2 involved data acquisition after urination. During Phase 1, we collected data for a total of 10 minutes each time the participant felt the urge to urinate. Wireless data points were captured every 6 seconds using the wearable prototype. To simulate daily movements and prevent over-fitting of the machine learning model, participants were instructed to walk approximately 5 steps between two consecutive data acquisitions.

Following Phase 1, the participant emptied her bladder while wearing the prototype. In Phase 2, the participant repeated the data acquisition process from Phase 1, but this time it was performed after urination. The same 10-minute data collection interval was followed, resulting in an additional set of data points. The collected data was then stored in CSV files and subjected to further analysis.

We performed statistical analysis to interpret the relationship between urination status and electrode potentials. Figure 5 illustrates a correlation heat map, where each square represents the Pearson correlation coefficient between two features. For example, the bottom line represents the correlation between the impedance values and pre, post urination statuses. The darker the square is, the less correlation between the two.

Additionally, we generated a violin plot to visualize the feature distribution against each pre urination and post urination statuses as shown in Figure 6. In Figure 6, the X-Axis represents the impedance of any two cross-sectional electrode pairs (for example, the 18 means the electrodes number 1 and number 8.), while the Y-Axis represents the bio-impedance value. By examining the plot, it becomes apparent that there is a notable disparity in the distribution of bio-impedance data between the two statuses of pre and post urination.

Next, machine learning analysis were conducted. The first two round experiment data were used as the training set while the last round experiment data were used as testing set. The results showed 100% accuracy. The results affirm the Privee's effectiveness in its intended purpose. By leveraging machine learning techniques, we have provided evidence supporting the reliability and utility of Privee.

5 CONCLUSION AND FUTURE DIRECTIONS

The key innovation of this study lies in the design of Privee, which incorporates eight embroidered electrodes and textile transmission lines seamlessly integrated into the fabric of the undergarment. Privee enables noninvasive and continuous monitoring of bladder fullness using bioimpedance spectroscopy technology, leveraging the electrical properties of body tissues and fluids. By analyzing the bioimpedance signals, Privee provides real-time information about the bladder's fullness level, offering a user-friendly and convenient solution for individuals with urinary incontinence. The noninvasive nature of Privee eliminates discomfort and potential risks associated with invasive bladder monitoring methods. With its potential to improve patients' quality of life and optimize healthcare costs, Privee demonstrates a promising approach to shift the focus from containment to proactive monitoring and warning systems for urinary incontinence management.

In the future, our research will emphasize the assessment of effectiveness through expanded human subject testing in clinical conditions involving a larger sample size and a greater number of tests. This will enable us to gather more comprehensive and robust data to strengthen the validity of our findings. Moreover, as additional data becomes available, our machine learning model will undergo continuous training, enhancing its accuracy and predictive capabilities. To further advance the practical applicability of our research, we aspire to collaborate with hospitals and initiate clinical trials for Privee. This collaborative effort will provide invaluable insights into the real-world implementation and potential benefits of our findings in a clinical setting.

ACKNOWLEDGMENTS

The authors would like to acknowledge the New Research and Scholarship Initiatives and Collaborative Interdisciplinary Research Grant of the University of California, Davis Division of the Academic Senate.

REFERENCES

- [1] Paul Abrams, Karl-Erik Andersson, L Birder, L Brubaker, Linda Cardozo, C Chaple, A Cottenden, W Davila, Dirk De Ridder, Roger Dmochowski, et al. 2010.

- Fourth International Consultation on Incontinence Recommendations of the International Scientific Committee: Evaluation and treatment of urinary incontinence, pelvic organ prolapse, and fecal incontinence. *Neurourology and Urodynamics: Official Journal of the International Continence Society* 29, 1 (2010), 213–240.
- [2] Brijesh Kumar Agarwal and Namita Agarwal. 2017. Urinary incontinence: prevalence, risk factors, impact on quality of life and treatment seeking behaviour among middle aged women. *International Surgery Journal* 4, 6 (2017), 1953–1958.
- [3] Lisa Beckmann, Dirk van Riesen, and Steffen Leonhardt. 2007. Optimal electrode placement and frequency range selection for the detection of lung water using bioimpedance spectroscopy. In *2007 29th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. IEEE, 2685–2688.
- [4] Andrea C Buchholz, Cynthia Bartok, and Dale A Schoeller. 2004. The validity of bioelectrical impedance models in clinical populations. *Nutrition in Clinical Practice* 19, 5 (2004), 433–446.
- [5] Antonino De Lorenzo, Angela Andreoli, James Matthie, and P Withers. 1997. Predicting body cell mass with bioimpedance by using theoretical methods: a technological review. *Journal of applied physiology* 82, 5 (1997), 1542–1558.
- [6] Quan Dong, R Scott Downen, Baichen Li, Nam Tran, and Zhenyu Li. 2021. A cloud-connected multi-lead electrocardiogram (ECG) sensor ring. *IEEE Sensors Journal* 21, 14 (2021), 16340–16349.
- [7] Carrie Earthman, Diana Traugbber, Jennifer Dobratz, and Wanda Howell. 2007. Bioimpedance spectroscopy for clinical assessment of fluid distribution and body cell mass. *Nutrition in Clinical Practice* 22, 4 (2007), 389–405.
- [8] Ruijie Fang, Ruoyu Zhang, Elahe Hosseini, Mahdi Orooji, Houman Homayoun, Sayed Mohammad Hosseini, Mahya Faghih, Soheil Rafatirad, and Setareh Rafatirad. 2022. Atlas: An adaptive transfer learning based pain assessment system: A real life unsupervised pain assessment solution. In *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. IEEE, 1331–1337.
- [9] Ruijie Fang, Ruoyu Zhang, Elahe Hosseini, Anna M Parenteau, Sally Hang, Setareh Rafatirad, Camelia E Hostinar, Mahdi Orooji, and Houman Homayoun. 2022. Prevent over-fitting and redundancy in physiological signal analyses for stress detection. In *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, 2585–2588.
- [10] Ruijie Fang, Ruoyu Zhang, Elahe Hosseini, Anna M Parenteau, Sally Hang, Setareh Rafatirad, Camelia E Hostinar, Mahdi Orooji, and Houman Homayoun. 2022. Towards Generalized ML Model in Automated Physiological Arousal Computing: A Transfer Learning-Based Domain Generalization Approach. In *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, 2577–2584.
- [11] Ruijie Fang, Ruoyu Zhang, Sayed M Hosseini, Mahya Faghih, Soheil Rafatirad, Setareh Rafatirad, and Houman Homayoun. 2022. Pain level modeling of intensive care unit patients with machine learning methods: An effective congeneric clustering-based approach. In *2022 4th International Conference on Intelligent Medicine and Image Processing*. 89–95.
- [12] Kieran Foley, Marie Keegan, Iain Campbell, Brian Murby, Duncan Hancox, and Brian Pollard. 1999. Use of single-frequency bioimpedance at 50 kHz to estimate total body water in patients with multiple organ failure and fluid overload. *Critical care medicine* 27, 8 (1999), 1472–1477.
- [13] Gozde Goncu-Berk. 2019. Design Parameters for a Wearable Urinary Incontinence Device. In *ITech Style Summit, International Conference on Textile and Clothing*. Porto: Portugal.
- [14] Gozde Goncu-Berk, Ruoyu Zhang, and Cigdem Yilmaz. 2021. CalmWear: A Smart Tactile Sensory Stimulation Clothing. In *Proceedings of the 2021 ACM International Symposium on Wearable Computers*. 184–188.
- [15] Steven B Heymsfield, Z Wang, Marjolein Visser, Dymrna Gallagher, and Richard N Pierson Jr. 1996. Techniques used in the measurement of body composition: an overview with emphasis on bioelectrical impedance analysis. *The American journal of clinical nutrition* 64, 3 (1996), 478S–484S.
- [16] Ahmad Keshkar, Asghar Mesbahi, and Parinaz Mehnati. 2008. The effect of bladder volume changes on the measured electrical impedance of the urothelium. *International Journal of Biomedical Engineering and Technology* 1, 3 (2008), 287–292.
- [17] Xuyang Li, Elena Pancera, Lukasz Niestoruk, Wilhelm Stork, and Thomas Zwick. 2010. Performance of an ultra wideband radar for detection of water accumulation in the human bladder. In *The 7th European Radar Conference*. IEEE, 212–215.
- [18] Henry C Lukaski. 1996. Biological indexes considered in the derivation of the bioelectrical impedance analysis. *The American journal of clinical nutrition* 64, 3 (1996), 397S–404S.
- [19] Andrew J Macnab and Lynn Stothers. 2008. Near-infrared spectroscopy: validation of bladder-outlet obstruction assessment using non-invasive parameters. *Can J Urol* 15, 5 (2008), 4241–4248.
- [20] James R Matthie. 2008. Bioimpedance measurements of human body composition: critical analysis and outlook. *Expert review of medical devices* 5, 2 (2008), 239–261.
- [21] O Contreras Ortiz. 2004. Stress urinary incontinence in the gynecological practice. *International Journal of Gynecology & Obstetrics* 86 (2004), S6–S16.
- [22] Jens Passauer, Helga Petrov, Andreas Schleser, Joachim Leicht, and Klaus Pucalka. 2010. Evaluation of clinical dry weight assessment in haemodialysis patients using bioimpedance spectroscopy: a cross-sectional study. *Nephrology Dialysis Transplantation* 25, 2 (2010), 545–551.
- [23] Lúcio T Santos, Martin Tygel, and Antonio C Bugginga Ramos. 2002. Reflection impedance. In *2002 SEG Annual Meeting*. OnePetro.
- [24] SL York, LC Ward, S Czerniec, MJ Lee, KM Refshauge, and SL Kilbreath. 2009. Single frequency versus bioimpedance spectroscopy for the assessment of lymphedema. *Breast cancer research and treatment* 117 (2009), 177–182.
- [25] Ruoyu Zhang, Ruijie Fang, Zhichao Zhang, Elahe Hosseini, Mahdi Orooji, Houman Homayoun, and Gozde Goncu-Berk. 2023. Short: Real-Time Bladder Monitoring by Bio-impedance Analysis to Aid Urinary Incontinence. In *2023 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*. IEEE, 138–142.
- [26] Zhichao Zhang, Ruoyu Zhang, Chi-Wei Chang, Yaojun Guo, Yung-Wei Chi, and Tingrui Pan. 2021. iwrap: A theranostic wearable device with real-time vital monitoring and auto-adjustable compression level for venous thromboembolism. *IEEE Transactions on Biomedical Engineering* 68, 9 (2021), 2776–2786.